# Worksheet 6.2 — Double Angle Formulas

Grade 12 Advanced Functions (MHF4U) · Unit 6: Trigonometric Identities and Equations

## The Big Ideas

Setting $B = A$: $\sin 2\theta = 2\sin\theta\cos\theta$ and $\cos 2\theta = \cos^2\theta - \sin^2\theta = 1 - 2\sin^2\theta = 2\cos^2\theta - 1$.

- $\sin 2\theta = 2\sin\theta\cos\theta$.
- $\cos 2\theta = 1 - 2\sin^2\theta = 2\cos^2\theta - 1$.
- $\tan 2\theta = \frac{2\tan\theta}{1-\tan^2\theta}$.

## Core Concepts

### Doubling an angle

Setting $B = A$ in the compound formulas yields the **double angle** identities, such as $\sin 2A = 2\sin A\cos A$.

### Cosine's three forms

$\cos 2A = \cos^2 A - \sin^2 A = 2\cos^2 A - 1 = 1 - 2\sin^2 A$ pick the version that fits your information.

### Simplifying

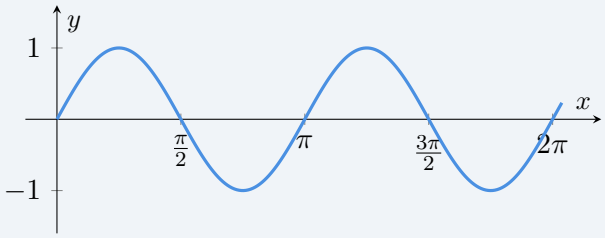
Double angle formulas condense expressions and are essential for proving identities and solving equations.

## Worked Examples

### Example 1 — Sine double angle

$\sin\theta = \frac{8}{17}, \cos\theta = \frac{15}{17}$. Find $\sin 2\theta$.

***Solution.*** $2 \cdot \frac{8}{17} \cdot \frac{15}{17} = \frac{240}{289}$. The graph $y = \sin 2x$ completes two cycles in $2\pi$:




### Example 2 — Cosine double angle

With $\sin\theta = \frac{2}{3}$, find $\cos 2\theta$.

***Solution.*** $1 - 2\sin^2\theta = 1 - \frac{8}{9} = \frac{1}{9}$.

### Example 3 — Recognize

Simplify $2\sin 3x\cos 3x$.

***Solution.*** $\sin 6x$.

**Example 4 — Cosine form**

Write $\cos 2\theta$ using only $\sin\theta$.

***Solution.*** $1 - 2\sin^2\theta$.

**Example 5 — Recognize**

Simplify $1 - 2\sin^2 4x$.

***Solution.*** $\cos 8x$.

**Example 6 — Sine double angle**

$\sin\theta = \frac{7}{25}, \cos\theta = \frac{24}{25}$. Find $\sin 2\theta$.

***Solution.*** $2 \cdot \frac{7}{25} \cdot \frac{24}{25} = \dfrac{336}{625}$.

**Example 7 — Recognize**

Simplify $2\cos^2 5x - 1$.

***Solution.*** $\cos 10x$.

**Example 8 — Cosine double angle**

With $\cos\theta = \frac{5}{13}$, find $\cos 2\theta$.

***Solution.*** $2 \cdot \frac{25}{169} - 1 = -\dfrac{119}{169}$.

**Example 9 — Simplify**

Simplify $\dfrac{\sin 2x}{2\sin x}$.

***Solution.*** $\dfrac{2\sin x\cos x}{2\sin x} = \cos x$.

## Your Turn — Show Your Work

For each question, show all steps clearly in the space provided.

### Question 1

$\sin\theta = \frac{9}{41}, \cos\theta = \frac{40}{41}$. Find $\sin 2\theta$.

### Question 2

Simplify $2\cos^2\frac{x}{2} - 1$.

### Question 3

With $\cos\theta = \frac{12}{13}$, find $\cos 2\theta$.

### Question 4

Simplify $\dfrac{\sin 2x}{\sin x}$.

### Question 5

Simplify $\cos^2 3x - \sin^2 3x$.

## Question 6

State $\sin 2\theta$.

## Question 7

Simplify $2\sin\frac{x}{2}\cos\frac{x}{2}$.

## Question 8

With $\sin\theta = \frac{3}{4}$, find $\cos 2\theta$ (use $1 - 2\sin^2\theta$).

## Question 9

Evaluate $1 - 2\sin^2 15°$ exactly.

## Question 10

State the three forms of $\cos 2\theta$.

## Question 11

With $\sin\theta = \frac{20}{29}, \cos\theta = \frac{21}{29}$, find $\sin 2\theta$.

## Question 12

Simplify $\dfrac{2\tan x}{1-\tan^2 x}$.

## Question 13 — Challenge

Given $\cos\theta = \frac{35}{37}$ ($\theta$ in QI), find both $\sin 2\theta$ and $\cos 2\theta$.

# Question Recap & Answer Key

## Question Recap

1. $\sin\theta = \frac{9}{41}, \cos\theta = \frac{40}{41}$. Find $\sin 2\theta$.
2. Simplify $2\cos^2\frac{x}{2} - 1$.
3. With $\cos\theta = \frac{12}{13}$, find $\cos 2\theta$.
4. Simplify $\dfrac{\sin 2x}{\sin x}$.
5. Simplify $\cos^2 3x - \sin^2 3x$.
6. State $\sin 2\theta$.
7. Simplify $2\sin\frac{x}{2}\cos\frac{x}{2}$.
8. With $\sin\theta = \frac{3}{4}$, find $\cos 2\theta$ (use $1 - 2\sin^2\theta$).
9. Evaluate $1 - 2\sin^2 15°$ exactly.
10. State the three forms of $\cos 2\theta$.
11. With $\sin\theta = \frac{20}{29}, \cos\theta = \frac{21}{29}$, find $\sin 2\theta$.
12. Simplify $\dfrac{2\tan x}{1 - \tan^2 x}$.
13. Given $\cos\theta = \frac{35}{37}$ ($\theta$ in QI), find both $\sin 2\theta$ and $\cos 2\theta$.

## ✓ Answer Key

1. $\frac{720}{1681}$
2. $\cos x$
3. $\frac{119}{169}$
4. $2\cos x$
5. $\cos 6x$
6. $2\sin\theta\cos\theta$
7. $\sin x$
8. $-\frac{1}{8}$
9. $\cos 30° = \frac{\sqrt{3}}{2}$
10. $\cos^2\theta - \sin^2\theta = 1 - 2\sin^2\theta = 2\cos^2\theta - 1$
11. $\frac{840}{841}$
12. $\tan 2x$
13. $\sin 2\theta = \frac{840}{1369}$, $\cos 2\theta = \frac{1081}{1369}$